

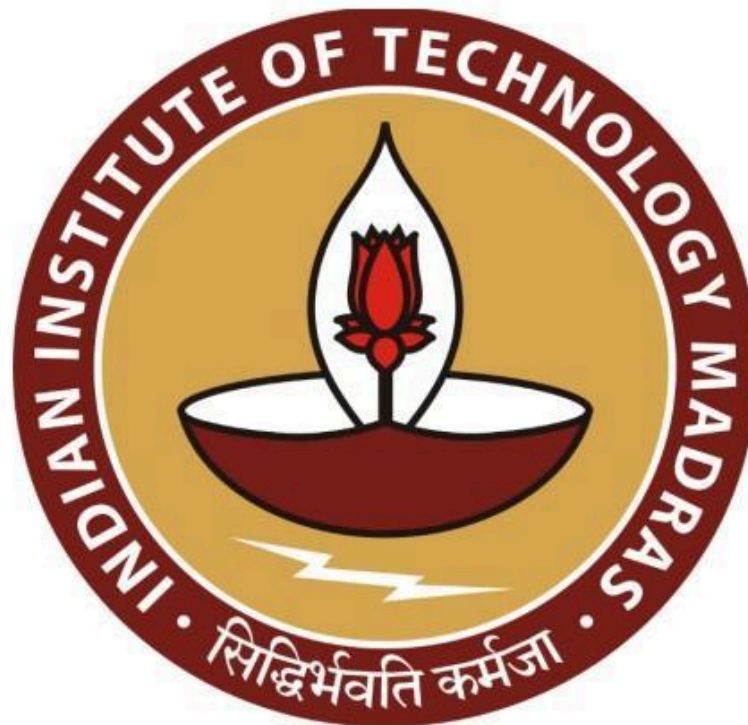
Improving Operational Efficiency in B2C Agribusiness: A Data-Driven Analysis for Sairanga Pesticides & Seeds

A Midterm report for the BDM Capstone Project

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1. EXECUTIVE SUMMARY

Sairanga Pesticides & Seeds, a B2C agribusiness retail outlet based in Kanaganapalli, Sri Sathya Sai District, Andhra Pradesh, specializes in providing essential agricultural inputs such as pesticides, fertilizers, biostimulants, and crop protection products to local farmers. Despite a strong customer base and a growing product portfolio, the business faces challenges in customer segmentation, inventory management, and product-wise profitability tracking. The absence of structured data analytics and forecasting mechanisms has limited efficient stock control, demand anticipation, and profit optimization, thereby impacting overall operational efficiency and growth potential in a competitive rural market.

To address these operational inefficiencies, detailed primary data was collected from August 2024 to July 2025. This dataset includes monthly product-wise sales, inventory stock levels, and pricing details recorded across multiple Excel sheets namely product, product-sales-month-wise, inventory, and sales. Key metadata such as Product ID, Product Name, Category, Cost Price, Selling Price, Revenue, and Profit were systematically organized and verified with the shop's billing and stock records.

The data was analyzed using Python tools such as Pandas, Matplotlib, and Seaborn for data cleaning, visualization, and computation. Analytical models like RFM Analysis, ABC Classification, and Holt Linear Trend Forecasting were applied to understand customer behavior, optimize inventory, and predict product demand. Additionally, Scenario Modeling was performed to evaluate the impact of price and cost changes on overall profitability.

2. PROOF OF ORIGINALITY

The data used in this report is primarily collected directly from Sairanga Pesticides & Seeds, a retail agribusiness outlet located in Kanaganapalli, Sri Sathya Sai District, Andhra Pradesh. A meeting was conducted with Mr. Rajesh, the owner of Sairanga Pesticides & Seeds, to confirm the accuracy of the data. A confidentiality agreement was signed to ensure proper use and secure handling of the information. The data includes customer-wise sales records, product-wise

monthly transactions, pricing information, and inventory stock details. The following documents can be accessed by using the following –[Link](#)

3. METADATA

The sales data used in this project was entered every day into Excel while selling products. It contains 2733 records and 10 columns. Each record shows one sale. The main details in this sheet are Product ID, Product Name, Category, Cost Price, Selling Price, Units Sold, Revenue, Profit, Customer Name, and the Date of Sale.

We also have an inventory dataset, which was prepared using monthly purchase bills and internal stock notes. This sheet has 40 rows and 63 columns, where each row represents a product and the columns show month-wise stock details. It includes Opening Stock, Quantity Sold, Incoming Stock, outgoing and Closing Stock for each month. They have also given Credit details which includes customer id, customer names and payment related details.

ATTRIBUTE	DATA_TYPE	DESCRIPTION
Product_ID	String	Unique ID of product.
Product_Name	String	Name of the product.
Category	String	Product category.
Cost_Price	Numerical	Buying price.
Selling_Price	Numerical	Selling price.
Profit_per_Unit	Numerical	Profit per unit.
Units_Sold	Numerical	Units sold.
Revenue	Numerical	Total sales.
Profit	Numerical	Total profit.
Month	Date	Sales month.
Opening_Stock	Numerical	Opening stock.
Closing_Stock	Numerical	Closing stock.
Dealer_Name / Customer_Name	String	Customer/Dealer name.
Transaction_Date	Date	Transaction date.
Bill_Amount	Numerical	Bill amount.
Payment_Status	String	Payment status.

Table 1: Meta Data

3.1 Data Cleaning Process: I collected the data from Sairanga Pesticides & Seeds and recorded it in Excel from August 2024 to July 2025. While cleaning the data, I corrected the missing cost and selling prices using the shop's product list. I also fixed spelling differences in product names, made the dates follow one format that is Indian format, and removed duplicate and empty rows from the given Excel sheet. After that, I matched and combined the product, sales, and stock sheets using Product ID and Product Name. Finally I have taken this data for my analysis by cross verifying with the owner of the shop.

4. DESCRIPTIVE STATISTICS

	YEAR_MONTH	TOTAL_REVENUE	MEAN_REVENUE	MEDIAN_REVENUE	MOST_SOLD_PRODUCT	LEAST_SOLD_PRODUCT
0	Aug-24	₹112,100	₹2,802	₹2,400	BLOOM	AMAZE Granules (Loose in Drum)
1	Sep-24	₹152,650	₹3,816	₹3,150	Sangamithra seeds	SAKTHI
2	Oct-24	₹201,100	₹5,028	₹4,325	VALENCIA ZINC EDTA 12%(500gm)	Craze
3	Nov-24	₹175,700	₹4,392	₹3,600	POINTER	AMAZE Granules (Loose in Drum)
4	Dec-24	₹106,350	₹2,659	₹2,400	POINTER	ACT
5	Jan-25	₹93,600	₹2,340	₹1,625	Gronior	NUTRI PLUS (FORMULA 6)
6	Feb-25	₹126,250	₹3,156	₹2,800	V-CARE	VALENCIA BORON 20%
7	Mar-25	₹224,350	₹5,609	₹4,300	V CONTROL PLUS	V-GREEN
8	Apr-25	₹221,950	₹5,549	₹3,325	AMAZE Granules (Bag)	V-CONTROL
9	May-25	₹124,750	₹3,119	₹2,675	SAAF	MENT PLUS
10	Jun-25	₹111,450	₹2,786	₹2,175	AMAZE Granules (Bag)	NUTRI PLUS (FORMULA 6)
11	Jul-25	₹147,150	₹3,679	₹2,725	NUTRI PLUS (FORMULA 6)	DiAmmonia phosphate

Table 2: Monthly Revenue Summary (Aug 2024 – Jul 2025)

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From August 2024 to July 2025, The month of March 2025 recorded the highest total revenue at ₹2,24,350 and January 2025 had the lowest total revenue at ₹93,600 indicating a slow sales period after harvest. Months like October 2024 and April 2025 also showed strong performance with revenues consistently above ₹2 lakh, while December 2024 and June 2025 displayed moderate sales levels.

In the peak period, especially March and April, products like V-Control Plus and Amaze Granules (Bag) were purchased more, indicating higher demand during crop preparation and protection phases and products such as Nutri Plus (Formula 6), V-Green, and Di-Ammonia Phosphate appeared frequently as the least sold products

	Category	COUNT	MEAN	MEDIAN	MODE	P25	P50	P75	MIN	MAX
0	Biostimulant	709	₹5,259.17	₹4,000	₹2,400	2600.000000	4000.000000	6612.500000	₹0	₹23,100
1	Fertilizer	442	₹3,184.52	₹1,775	₹4,400	1200.000000	1775.000000	4250.000000	₹0	₹16,800
2	Fungicide	264	₹5,201.04	₹4,200	₹2,000	2475.000000	4200.000000	7625.000000	₹500	₹16,000
3	Herbicide	428	₹3,550.69	₹2,550	₹750	1500.000000	2550.000000	5100.000000	₹0	₹13,600
4	Insecticide	544	₹2,755.73	₹2,400	₹1,000	1500.000000	2400.000000	3925.000000	₹200	₹7,200
5	Micronutrient	275	₹1,355.21	₹1,200	₹800	800.000000	1200.000000	1850.000000	₹200	₹4,550
6	Seeds	71	₹5,325.00	₹5,400	₹5,400	4275.000000	5400.000000	7200.000000	₹0	₹9,900

Table 3: Category-Wise Revenue Summary (Aug 2024 – Jul 2025)

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Based on the category-wise revenue summary, Biostimulants were sold the most, with 709 sales and an average revenue of ₹5,259.17. This shows that farmers buy these products regularly throughout the year. Seeds also have a high average revenue of ₹5,325.00, but the sales count is low, which means seeds are costly products and farmers buy them only when required. Fungicides also have a good average revenue of ₹5,201.04, which shows they are important during crop protection times.

Herbicides and fertilizers show steady and regular usage with average revenues of ₹3,650.69 and ₹3,184.52, meaning farmers need these throughout the farming season. Insecticides have a lower average revenue of ₹2,755.73, showing they are used but mostly in smaller amounts. Micronutrients have the lowest average revenue of ₹1,355.21, which means these are either lower priced or needed only in specific situations when crops need nutrient support.

5. DETAILED EXPLANATION OF ANALYSIS PROCESS/METHOD

5.1 Data Loading and Initial Analysis: The data was collected from Sairanga Pesticides & Seeds and recorded in Excel. This data included product details, monthly sales, inventory stock, and customer purchase information. These files were loaded into Python using Pandas to make analysis easier. The main columns used were Product ID, Product Name, Category, Cost Price, Selling Price, Units Sold, Revenue, and Profit. This helped in understanding how products were selling and how stock was changing every month.

5.2 Data Preprocessing: Missing values in cost and selling price were corrected by checking the shop's product list. Product names were standardized to avoid duplicates. Dates were formatted in one format. The product sheet, sales sheet, and inventory sheet were combined using Product ID. After cleaning, the data became consistent and ready for analysis.

5.3 Descriptive Statistics: Monthly revenue and product sales were summarized. The results showed that March 2025 had the highest revenue, while January 2025 had the lowest. Biostimulants were the most frequently purchased products, and Seeds had fewer sales but higher value per sale. This helped in understanding which products sell more and during which months.

5.4 Customer Segmentation (RFM Analysis): RFM analysis was used to group customers based on how recently they purchased, how often they purchased, and how much they spent. Customers were grouped into Platinum, Gold, Growth Potential, Occasional, and At Risk categories. This helps in deciding which customers should be contacted more often and which customers need to be re-engaged.

5.5 Product Level Sales Analysis: An ABC classification was performed to analyze product sales based on units sold, categorizing products into A, B, and C groups. This analysis provided insights into product demand, helping to prioritize high-performing products and optimize inventory management. Additionally, a SARIMA model was used to forecast future sales, assisting in better resource allocation and production planning. A Holt Linear Trend forecasting model was applied to predict product-wise sales for the next 3 months.

5.6 Profitability and Scenario Analysis: Profitability was calculated for each product using the difference between cost price and selling price. A Product-Wise Profit Bar Chart showed that products like Gronior, VN PLUS, and ACT gave the highest profit, while some micronutrient products gave very low profit. A scenario test was done by slightly increasing selling price and reducing cost price, and a Profit Comparison Chart showed that profit improved for selected products. This helps decide which products can give better returns with small pricing adjustments.

6. RESULTS AND FINDINGS

6.1 Customer Segmentation Analysis: Gold Customers form the highest portion of the customer base, showing consistent and loyal purchasing behavior. Growth Potential and Platinum Customers also make up a significant share, indicating strong opportunities to convert them into high value repeat buyers. Occasional Customers purchase irregularly and can be targeted with seasonal reminders. At Risk and Dormant Customers show reduced or very low activity, suggesting the need for re-engagement strategies such as follow-up calls or small discount offers.

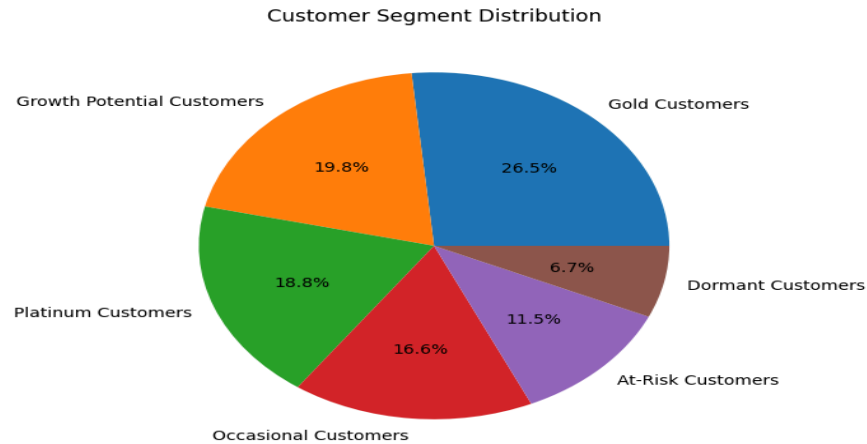


Figure 6.1.1:Pie Chart

Platinum and Gold Customers contribute the highest share of total revenue, showing that loyal customers are the main source of income for the shop. Growth Potential Customers provide a moderate revenue share and can be developed further through personalized support. Occasional Customers add limited revenue due to inconsistent buying patterns, while At-Risk and Dormant Customers contribute very little, highlighting the importance of improving retention and preventing customer drop-off.

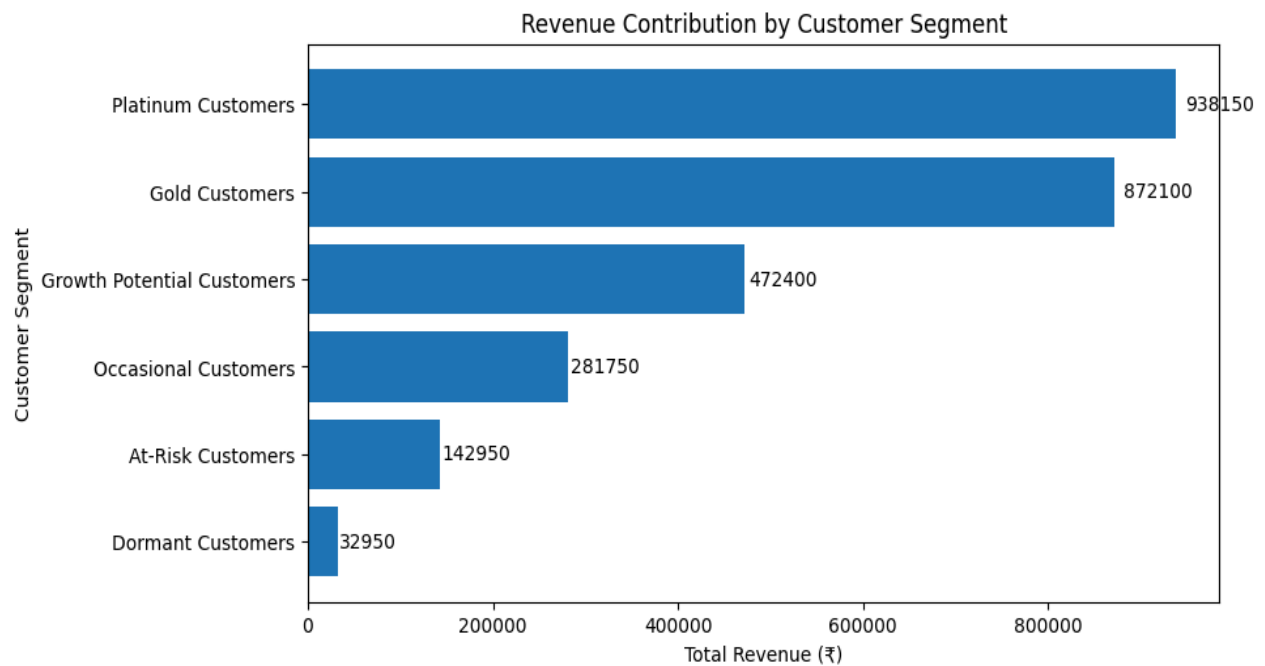


Figure 6.1.2:bar Chart

6.2 Product Level Demand Insights: The Monthly Sales: Actual vs. Predicted graph compares actual sales (bar chart) with sales predicted using the SARIMA (Seasonal ARIMA) model (red line).

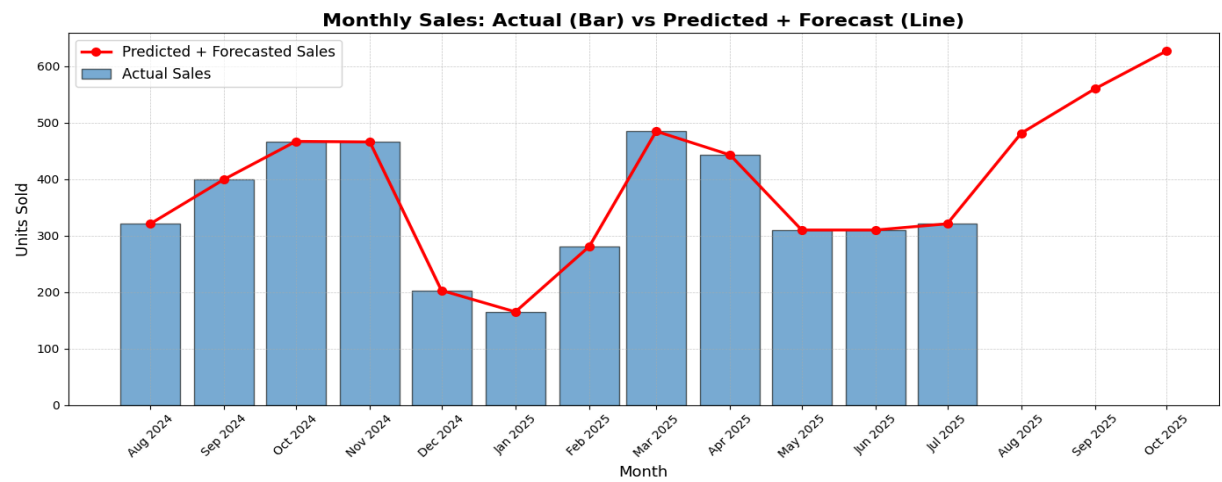


Figure 6.2.1: Monthly Sales: Actual vs. Predicted

The ABC Classification of Products Based on Units Sold graph categorizes products into high (green), moderate (orange), and low (red) sales segments. Category A products like P023 and P039 are the highest selling and should be prioritized for stock and production. Category B products show moderate demand and need controlled inventory. Category C products like P035 and P019 have low sales and may require promotion or reduction in stocking.

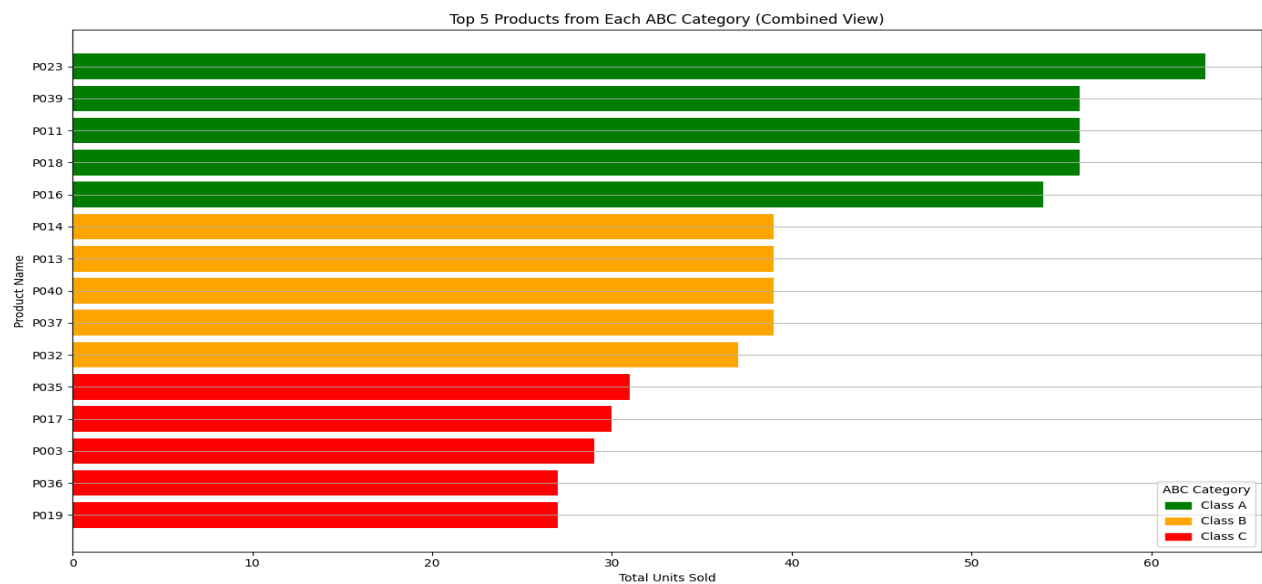


Figure 6.2.2: ABC Classification

The SARIMA model accounts for trends and seasonality in forecasting and I have forecasted for upcoming 3-months. I have also performed Holt's Linear Trend Model used to handle product-wise forecasting. In the SARIMA (Seasonal ARIMA) model, the forecast for the upcoming three months shows an upward movement because the model identified a rising trend in the sales pattern from the previous year. Since SARIMA learns from past seasonal and trend behavior, the consistent growth observed in earlier months influenced the model to project a similar increase in the future.

6.3 Product Wise Margin Analysis for Profitability Improvement: The analysis shows that most of the company's profit comes from only a few products. Gronior, VN PLUS, and ACT are the top profit makers, and Biostimulants as a category generate the highest earnings, followed by Fungicides and Insecticides. On the other hand, Micronutrients and Seeds contribute very little, so their pricing or stock levels may need to be reconsidered.

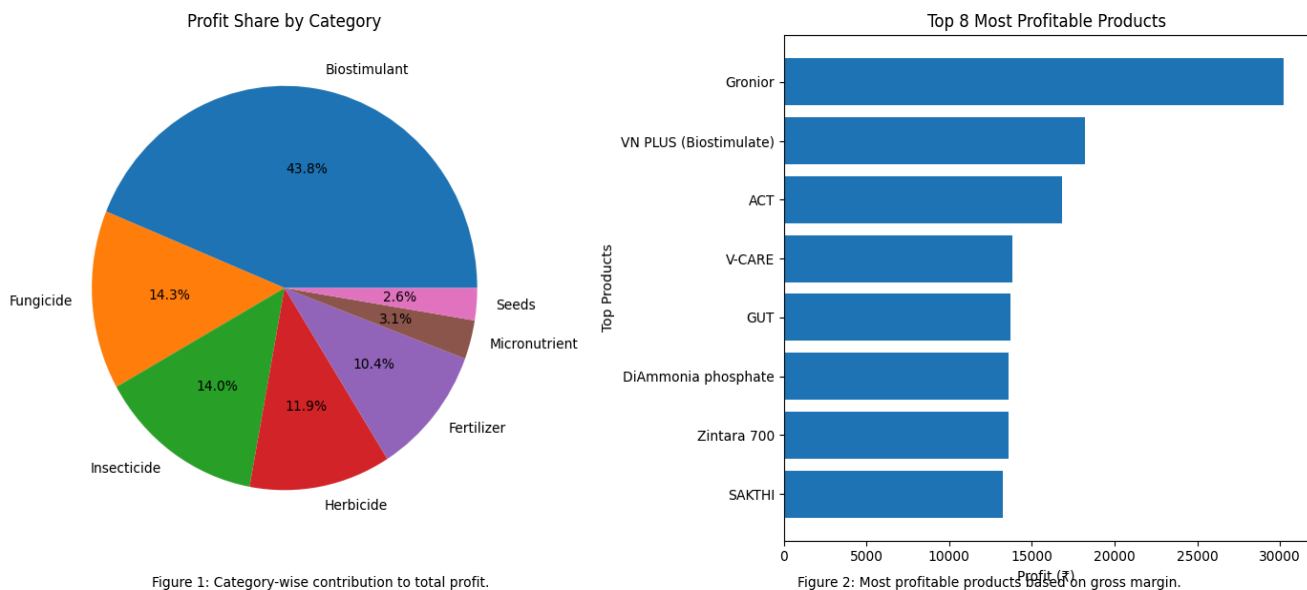


Figure 6.3.1 & 6.3.2: Category-wise contribution to total profit and Most profitable products based on gross margin

The scenario test was performed on the first top ten products, where the selling price was increased by 10% and cost reduced by 10%, showing a strong rise in profit for products like Zintara 700, AG ONE, and Predator. However, products like Pendamil Super did not show much improvement, which means their cost structure needs to be reviewed.

Overall, focusing more on high-profit products and adjusting pricing strategies can help the company improve profits and grow sustainably.

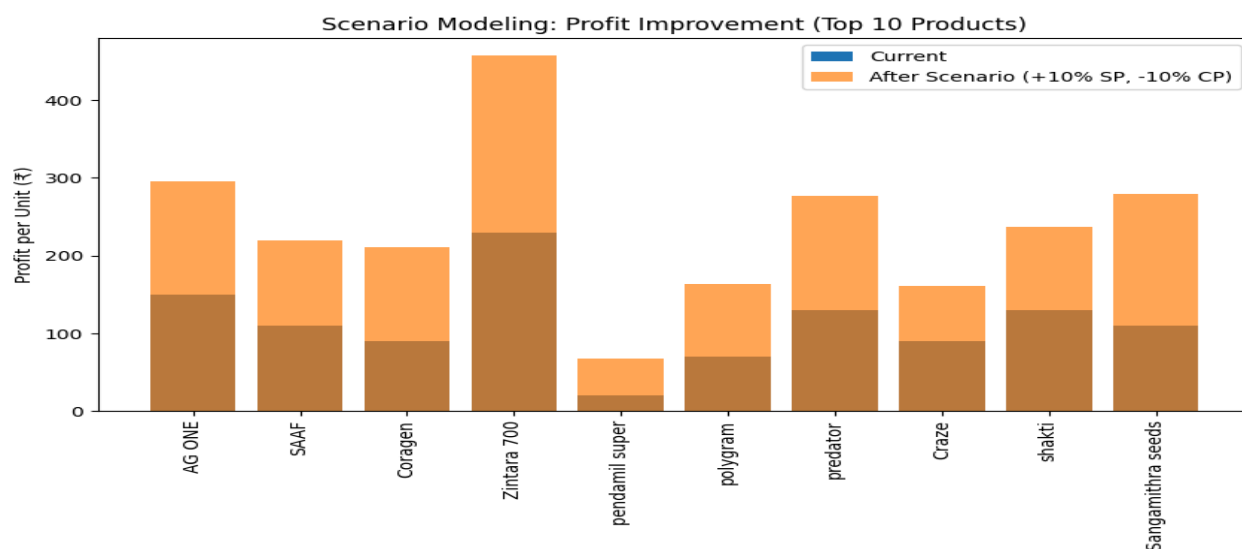


Figure 6.3.3: Scenario Modeling

7. Conclusion

This midterm analysis of Sairanga Pesticides & Seeds highlights important patterns in product demand, customer behavior, inventory movement, and profitability. The ABC classification showed that a small group of products contributes to most of the sales, meaning these high-demand items should be stocked regularly. Seasonal sales patterns and the SARIMA and Holt forecasting results indicated that demand increases during crop preparation months, helping in planning stock purchases more accurately. RFM analysis showed that loyal and regular customers contribute the highest revenue, while some customers require re-engagement to increase their activity. Profit analysis revealed that products like Gronior, VN PLUS, and ACT generate high margins, while some micronutrient items offer very low returns. The pricing scenario test demonstrated that small pricing adjustments can improve profit for selected products. Overall, the study shows that data driven decisions in stocking, pricing, and customer follow up can improve business efficiency and profitability.